

A novel hybrid model for forecasting crude oil price based on time series decomposition

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HIGHLIGHTS

- A novel hybrid model is proposed in order to forecast crude oil price.
- Time series is decomposed by complete ensemble empirical mode decomposition.
- Hybrid model based on artificial intelligence is used to predict nonlinear series.
- Hybrid model based on econometrics is employed to predict volatile components.
- Final forecast is the summation of the constituent models that outperforms counterparts.

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ABSTRACT

Oil price forecasting has received a prodigious attention by scholars and policymakers due to its significant effect on various economic sectors and markets. Incentivized by this issue, the author proposes a novel hybrid model for crude oil price forecasting whose focus is on improving the accuracy of prediction taking into consideration the characteristics existing in the oil price time series. In so doing, the author constitutes a hybrid model consisting of complete ensemble empirical mode decomposition, support vector machine, particle swarm optimization, and Markov-switching generalized autoregressive conditional heteroskedasticity to capture the non-linearity and volatility of the time series more effectively. Mean absolute error, root mean square error, and mean absolute percentage error tests are used to measure forecasting errors. Results robustness and forecasting quality of the proposed hybrid model compared with counterparts are also investigated by Diebold-Mariano test. Finally, empirical results demonstrate that the proposed hybrid model outperforms other models.

1. Introduction

Oil price prediction has been a striking issue in the field of forecasting research. Oscillations in oil prices significantly affect the economy of both oil-exporting and oil-importing nations [1]. For instance, a rise in oil price may cause inflation, and subsequently recession, in the oil-importer countries while a fall may hinder the economic development of the exporting countries [2]. Thus, a reliable forecast of oil price has become a pivotal issue which received a plethora of attention by all governments as it would help policy makers adopt proper policy and make appropriate decision regarding energy resources [3]. Also, research shows that oil price has a prodigious impact on other financial markets [1]. Motivated by this issue, the author aims to propose a novel hybrid model for crude oil price forecasting whose focus is on improving the accuracy of prediction considering the characteristics existing in the time series.

Overall, forecasting approaches can be categorized in three classes. First class encompasses statistical models such as autoregressive integrated moving average (ARIMA) [4], generalized autoregressive conditional heteroskedasticity (GARCH) [5], random walk (RW) [6], etc. Second class concerns with methods derived from artificial intelligence (AI) and expert systems like support vector machine (SVM) [7], adaptive neuro fuzzy inference system (ANFIS) [8], genetic algorithm (GA) [9], etc. Finally, the hybridization of the aforementioned models forms the last class [10]. Literature demonstrates that using hybrid models results in a more reliable forecast [11] as it improves the model ability in capturing various characteristics existing in the time series [12]. However, the statistical models effectively focus on the linearity and volatility of time series [13], but fail to capture the non-linearity and non-stationarity existing in the oil price time series [14]. Therefore, the forecasting accuracy of the statistical models subsides due to the very shortcomings. Research suggests using models derived

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Nomenclature

ARIMA	Autoregressive integrated moving average
AI	Artificial intelligence
ANFIS	Adaptive neuro fuzzy inference system
ARFIMA	Autoregressive fractionally integrated moving average
ARMA	Autoregressive moving average
ADF	Augmented Dicky-Fuller
CEEMD	Complete ensemble empirical mode decomposition
DM	Diebold-Mariano
EEMD	Ensemble empirical mode decomposition
EMD	Empirical mode decomposition

GA	Genetic algorithm
GARCH	Generalized autoregressive conditional heteroskedasticity
IMF	Intrinsic mode function
MAE	Mean absolute error
MAPE	Mean absolute percentage error
MS	Markov-switching
OPEC	Organization of the Petroleum Exporting Countries
RMSE	Root mean square error
RW	Random walk
SVM	Support vector machine
WTI	West Texas Intermediate

from AI in order to deal with the abstruse characteristics of oil price time series [15]. However, Makridakis et al. [16] compare the forecasting accuracy of the AI models with that of statistical models and report that statistical models outperform their counterparts in some cases, especially when the linear relationship is more powerful in the time series. Owing to the simultaneous existence of various characteristics in the oil price time series, i.e. linearity, volatility, lag, structural breaks, nonlinearity, etc., using a combination of the statistical and AI models received a plethora of attention by scholars. According to the literature, the forecasting accuracy directly depends on capturing the characteristics existing in the time series [17]. Previous studies applied hybrid models consisting of different statistical and AI techniques using weighting functions to assign an optimal weight to each technique in order to optimize the power of hybrid models in capturing these characteristics. Although this approach led to more promising results, it is, in essence, a type of integrated approach where, for instance, a statistical approach designed to capture linear relationships plays a part in forecasting a complex nonlinear time series which accordingly has a negative effect on the final prediction accuracy. An effective medium to overcome this shortcoming is to utilize the so-called “divide and conquer” strategy in the forecasting realm. That is to say, by decomposing the complex signal of the price time series into separate specific components for which the main characteristics can be easily identified, we then can adopt a particular forecasting model for each component based on the extracted features. Therefore, the objective of the proposed model is to decompose the oil price time series into volatile and nonlinear categories and utilize appropriate forecasting techniques for each component. However, it is not the first work where a decomposition method is applied to oil price time series. In retrospect, Wu et al. [18] proposed a hybrid model consisting of the complete ensemble empirical mode decomposition (CEEMD), ARIMA, and sparse Bayesian learning to forecast oil price, but they only used root mean square error (RMSE) for performance evaluation and suggested using more suitable forecasting techniques. In this paper, the author not only utilizes more tests for evaluation, namely mean absolute error (MAE), mean absolute percentage error (MAPE), and Diebold-Mariano (DM) tests, but also selects more appropriate techniques. ARIMA models generally outperform in capturing lags while GARCH models effectively capture time-varying volatility [19] which is a ubiquitous feature when it comes to oil price time series. The author hypothesizes that the proposed model outperforms similar counterparts and elevates the forecasting accuracy for the oil price.

Studies on oil price forecasting demonstrate that the prediction is sensitive to the nonlinearity, structural breaks, and high-volatile behavior of the time series. Thus, the author forms a hybrid model consisting of CEEMD, SVM, particle swarm optimization (PSO), and Markov-switching (MS) GARCH in an attempt to achieve a more accurate forecast compared to other methods with respect to the empirical analysis. Decomposing the oil price time series generates sub-series or components with nonlinear and volatile characteristics; therefore, the aim is to forecast these features through econometric

models and AI techniques separately. The merit of the proposed model lies in the fact that it forecasts either nonlinear feature or volatile feature through techniques which are efficient in capturing those characteristics. That is to say, the nonlinear and volatile components are forecasted by the SVM-PSO and MS-GARCH, respectively. Then, the linear addition of these forecasts gives us a more reliable estimation of the oil price. To show the efficiency of the proposed model, the author uses the data of West Texas Intermediate (WTI) crude oil prices and compared the results with those of individual models and also another state-of-the-art forecasting model already introduced in the literature.

The effectivity of the CEEMD makes it a useful tool to enhance the forecasting quality by dividing the original time series into several components or intrinsic mode functions (IMFs) based on their characteristics. Comparing with other decomposition algorithms, the merit of CEEMD is its ability to eliminate the residue noise. Historically, CEEMD is built on the empirical mode decomposition (EMD) which is an adaptive technique proposed to examine non-linear signals. However, using the EMD was accompanied by problems, like the existence of vacillations with egregious amplitudes in a mode or the existence of fairly identical vacillations in disparate modes (mode mixing). Later, ensemble empirical mode decomposition (EEMD) was introduced in order to tackle these shortcomings [20]. Taking advantage of Gaussian white noise caused the EEMD to overcome the so-called mode mixing problem but generated new problems such as residual noise for the reconstructed signals. In a seminal paper Torres et al. [21] proposed the CEEMD which is able to reconstruct the original signal and produce precise mode separation more effectively. Also, in comparison with similar counterparts like wavelet transform, the CEEMD is able to apprise soft changes in time-varying frequency. Thus, the author employs the CEEMD for data decomposition in the proposed model. By decomposing the data, I then forecast each component using a technique which is of the capacity to capture the corresponding features properly. For the nonlinear components, the SVM seems to be a successful method as it mitigates various shortcomings of alternative methods such as artificial neural network in terms of local convergence [22]. However, the paramount performance of SVM models is highly dependent on the proper selection of parameters. Therefore, the PSO is employed to ascertain optimal parameters for the SVM. PSO is a heuristic algorithm which investigates the search space to discover the best set of parameters. On the other hand, the author uses MS-GARCH to forecast volatile components derived from the CEEMD. In this approach, all parameters switch between low-volatile and high-volatile regimes while these regimes have disparate parameters. In addition to appraising low and high volatility of the oil prices, this method reduces skewness, kurtosis, and tail-fatness for volatility models [23].

The contribution of this paper to the literature is introducing a new model for a reliable oil price forecasting where the validation of the proposed model also presents more robust results compared with its counterparts as demonstrated by the DM test. The cardinal advantage of the proposed hybrid model is its capability to capture various features existing in the oil price time series through the particular characteristics

of the constituent models. Also, MS-GARCH characteristics in this study gives rise to a remarkable pliability in modeling the regime switches as well as persistence. The remaining parts of the paper are organized as follows. Section 2 presents a literature review of the oil price forecasting. Section 3 describes the applied methodology and formulation of the hybrid model. Section 4 describes the data and empirical results. Section 5 concludes the paper.

2. Literature review

Oil price forecasting has been an important area of research where numerous techniques and models have been proposed by researchers in an attempt to improve the accuracy of prediction. Amano [24] conducted the first research in the field of forecasting oil price using a small-scale econometric model. Tang and Hammoudeh [25] employed a nonlinear regression model to forecast the oil price of organization of the petroleum exporting countries (OPEC). Moshiri and Foroutan [26] used a hybrid model consisting of GARCH and ARIMA to predict oil futures prices. Also, Yu et al. [13] proposed an EMD-neural network ensemble learning approach for a more precise prediction of the oil price. Gori et al. [27] employed an ANFIS model to forecast monthly oil prices. Khashman and Nwulu [28] forecasted the oil price using back-propagation and SVM models and compared the results obtained. He et al. [29] applied the wavelet transform to elevate the accuracy of the oil price forecasting. Zhang et al. [30] built a hybrid model based on the EEMD and used AI techniques to forecast crude oil prices. Also, Chiroma et al. [31] introduced a forecasting approach for oil prices based on the GA and neural networks and proved the superiority of the proposed approach over alternative models. Zhao et al. [32] introduced a hybrid model combining deep learning and ensemble learning to forecast oil prices. Abdollahi and Ebrahimi [17] proposed a new hybridization by incorporating the ANFIS, MS, and autoregressive fractionally integrated moving average and an optimization technique to provide a reliable prediction for oil prices. Lu et al. [33] developed a dynamic Bayesian structural model using the spike-slab regression to forecast oil prices and identify turning points of the price. Lin et al. [34] compared the forecasting ability of GARCH-type models for the prediction of oil prices and proposed a hidden MS exponential GARCH model.

Some recent studies on oil price forecasting applied more sophisticated approaches. Zhang, Ma, Shi and Huang [35] introduced an

iterative hybrid model using macroeconomic variables in addition to technical indicators and verified the superiority of the proposed model over standard models. Tang et al. [36] proposed a non-iterative decomposition-ensemble learning approach using random vector functional link for oil price prediction. Their proposed model outperforms its counterparts in terms of forecasting accuracy. Bisoi et al. [37] proposed a robust random vector functional link network in which the variational mode decomposition is utilized to extract components from oil price time series. Tang et al. [38] presented a multi-scale model with search engine data using oil-related factors as predictors. They investigated the multi-scale relationship between oil price and search engine data and proved the superiority of their model over alternative methods through an empirical study. Moreover, using predictive regressions with a robust loss and regularization constraints, Hao et al. [39] improved the oil price forecasting accuracy compared with that of standard models. Lin et al. [40] proposed a hybrid model consisting of the wavelet de-noising, EMD, autoregressive moving average, and fractionally integrated GARCH techniques to forecast the oil price and verified its superiority over traditional methods. Furthermore, Table 1 summarizes the recent literature of oil price forecasting.

To be in line with the literature developed so far, the present paper aims to forecast the WTI crude oil price through a hybrid model designed to capture the characteristics of the oil price time series more efficiently. Decomposing the WTI oil price time series brings about sub-series with nonlinear and volatile features. Therefore, the proposed model uses different techniques to forecast each series with respect to its fundamental nature. This leads to a more reliable prediction where the linear addition of the forecasts provides an outcome with more accuracy compared to the results obtained by other methods.

3. Constituent and proposed hybrid models

3.1. CEEMD

CEEMD is a tool that decomposes data based on the time scale. It was built on the EMD introduced by Torres, Colominas, Schlotthauer and Flandrin [21]. As an adaptive technique, the EMD has the capacity to decompose a time series into various IMFs. Following, the EMD process is described.

- i. Specify all local minima and local maxima in time series

Table 1
Summary of the recent studies on oil price forecasting.

Literature	Models	Main results
Herrera et al. [41]	Random forests and neural networks compared with traditional econometric models	Forecasting results by the machine learning models are more reliable and can show the turning points.
Li et al. [42]	Hybrid model combining improved CEEMD with adaptive noise and ridge regression	The superiority of the proposed model over statistical models is verified. Using machine learning techniques noticeably enhances prediction quality.
Di Sanzo [43]	Simultaneous long memory and MS	Forecasting results within various time periods show more accurate prediction compared to results achieved by the selected GARCH models.
Wu et al. [44]	EEMD and long short-term memory	Results show that the proposed hybridization works well even with varying number of decomposition results.
Cen and Wang [45]	Long short-term memory based on prior knowledge data transfer	Results indicate that forecasting accuracy is significantly elevated by using data transfer when it comes to long short-term memory.
Elaziz et al. [46]	Hybrid model consisting of the ANFIS, GA, and modified salp swarm algorithm	Results demonstrate that the proposed hybrid model outperforms its counterparts, like PSO-ANFIS and ANFIS based on grey wolf optimization in terms of some forecasting accuracy.
Zhu et al. [47]	EEMD and optimal combined forecasting model	Forecasting results demonstrate the superiority of the proposed hybrid approach over its benchmark models such as Holt's exponential smoothing, general regression neural network, nonlinear autoregressive neural network, etc.
Huang and Wang [48]	A novel random wavelet neural network	Empirical results show that the proposed model outperforms benchmark methods like back propagation neural network, SVM, and wavelet neural network in terms of forecasting accuracy.
Ding [49]	A hybrid model consisting of the EEMD and artificial neural network	The author used information criteria as the model selection rule for the prediction process. The proposed model outperforms the counterparts.
Wang et al. [50]	A hybrid model consisting of data fluctuation network, back propagation neural network, and extreme learning machine	Results indicate that the proposed models outperform the constituent models in terms of accuracy

- ($x_{(i)}, t = 1, 2, \dots, T$);
- Connect all local maxima and local minima to form upper envelopes ($x_{up}(t)$) and lower envelopes ($x_{low}(t)$);
 - Calculate the local mean (m_i);
 - Extract the first IMF ($IMF_1(t) = x_i - m_i$);
 - Extract the first residue ($R_1(t) = m(t)$). If find more than two local extrema for $R_i(t)$ ($i = 1, 2, \dots, n$) return to ii.

Having analogous signal parts in the same position in different IMFs was a shortcoming of the EMD. Thus, Huang, Shen and Long [20] proposed the EEMD to mitigate the effect of the so-called mixing problem. They repeatedly carried out the EMD on the time series with incorporated white noises. The new time series is formulated as shown in Eq. (1).

$$x^i(t) = x(t) + w^i(t) \quad (1)$$

where x^i is the new time series, $x(t)$ denotes the underlying time series, and w^i represents the i^{th} white noise. The pertinent $IMF_k^i(t)$ is obtained by decomposing the $x^i(t)$. However, the EEMD uses the average of IMFs to ascertain the value of k which results in including residual noises for the recovered signal. CEEMD was proposed to solve this shortcoming [21]. Accordingly, the value of k for IMF and residue can be calculated as:

$$\bar{IMF}_k = i^{-1} \sum_{i=1}^I \left(IMF_1(r_{k-1}(t) + \varepsilon_{k-1} IMF_1(w^i(t))) \right) \quad (2)$$

$$r_k(t) = r_{k-1}(t) - \bar{IMF}_k \quad (3)$$

3.2. SVM

SVM was introduced by Boser et al. [51] in an attempt to reduce the classification error in training data. Later, a developed version of SVM was proposed by Cortes and Vapnik [52] based on the statistical learning theory. SVM is highly applicable in problems like classification and clustering. Training set of SVM is denoted by $(x_1, y_1), \dots, (x_m, y_m) \in N \times R$, where x is input data. In regression $\varepsilon - SV$, the objective is to find function $f(x)$ which has a maximum deviation of ε from actual value (y_i) for all input data. The linear function ($f(x)$) is formulated as shown in Eq. (4).

$$f(x) = \langle w, x \rangle + bw \in N, b \in R \quad (4)$$

where $\langle \rangle$ denotes interior multiplication and w is the smoothing parameter. That is to say, the value of Euclidean norm $\|w\|^2$ must be minimized to achieve the smoothness. Therefore:

$$\begin{cases} y_i - \langle w, x_i \rangle - b \leq \varepsilon \\ \langle w, x_i \rangle + b - y_i \leq \varepsilon \end{cases} \quad (5)$$

The above optimization problem (Eq. (5)) can be solved by determining $f(x)$, (x_i, y_i) , and ε . Adding ξ_1^* and ξ_i for all the unsolvable variables, the optimization problem is achieved:

$$\text{Minimize } 1/2\|w\|^2 + c \sum_{i=1}^I (\xi_1^* + \xi_i) \quad (6)$$

$$\begin{cases} y_i - \langle w, x_i \rangle - b \leq \varepsilon + \xi_i \\ \langle w, x_i \rangle + b - y_i \leq \varepsilon + \xi_i^* \\ \xi_1^*, \xi_i > 0 \end{cases} \quad (7)$$

Loss function ε - intensive is defined as Eq. (8).

$$|\xi_i| = \begin{cases} 0 & \text{if } |\xi_i| < \varepsilon \\ |\xi_i| < \varepsilon & \text{Otherwise} \end{cases} \quad (8)$$

Lagrangian Duality is a proper tool in order to enable the SVM to capture nonlinearity. Dual standard method uses Lagrangian

coefficients as shown in Eq. (9). Of note is that Eq. (9) is solved based on Smola and Schölkopf [53].

$$L = \frac{1}{2\|w\|^2} - \sum_{i=1}^I a_i(\varepsilon + \xi_i - y_i - \langle w, x_i \rangle + b) - \sum_{i=1}^I a_i^*(\varepsilon + \xi_1^* - y_i - \langle w, x_i \rangle - b) - \sum_{i=1}^I (y_i \xi_i + y_i^* \xi_1^*) + c \sum_{i=1}^I (\xi_i + \xi_1^*) \quad (9)$$

3.3. PSO

Kennedy and Eberhart [54] introduced the PSO as an optimization method for nonlinear functions. Numerous studies employed this algorithm due to its search efficiency and simplicity [55]. PSO has a population of N swarm particles within the search space where each swarm particle is positioned randomly and has its own velocity. At the beginning, some swarm particles are randomly positioned in the problem space trying to find the optimal solution by iteration and with respect to the optimization criteria. At each iteration, the particles are updated by two positions: First, the personal historical best position achieved by the particle is updated; second, the best position found by the total population is updated. Therefore:

$$v_j^i[t+1] = w * v_j^i[t] + C_1 * f_1(x_j^{i,best}[t] - x_j^i[t]) + c_1 * r_2(x_j^{g,best}[t] - x_j^i[t]) \quad (10)$$

$$x_j^i[t+1] = x_j^{i,best}[t] + v_j^i[t+1] \quad (11)$$

where d denotes the problem dimensions. x_j^i and v_j^i are respectively the position and velocity of the i^{th} particle at the j^{th} iteration. $x_j^{i,best}$ denotes the best position achieved by the i^{th} particle [56].

3.4. MS-GARCH

Bollerslev [57] proposes the GARCH model by developing autoregressive conditional heteroskedastic model. GARCH is of the capacity to successfully model the time-varying volatility in time series [58]. This model is formulated as:

$$h_t = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 h_{t-1} \quad (12)$$

where h_t represents the conditional variance at time t , ε_{t-1}^2 denotes the delayed residuals squared, and α_0 is the constant term.

Hamilton [59,60] proposes the MS method to statistically recognize the turning points existing in the time series. Accordingly, the model parameters are allowed to switch between regimes with respect to markov process. A two-regime markov chain is formulated as shown in Eq. (13).

$$P\{s_k=j|s_{k-1}=i, s_{k-2}=k, \dots\} = P\{s_k=j|s_{k-1}=i\} = p_{ij} \quad (13)$$

where s_k denotes the change in regimes, p_{ij} represents the probability of transition from regime i to regime j . Following a two-regime markov chain, the probabilities are:

$$\begin{aligned} Pr\{s_k=0|s_{k-1}=0\} &= p \\ Pr\{s_k=1|s_{k-1}=1\} &= q \\ Pr\{s_k=0|s_{k-1}=1\} &= 1-q \\ Pr\{s_k=1|s_{k-1}=0\} &= 1-p \end{aligned} \quad (14)$$

where $s_k = 0$ is the regime with normal volatility implying a low-volatile behavior while $s_k = 1$ denotes the jump regime with a high-volatile behavior. Gray [61] introduces the two-regime MS-GARCH as follows.

$$h_t = [\alpha_0 + \alpha_{1(s_t)} \varepsilon_{t-1}^2 + \beta_{1(s_t)} h_{t-1}] I[s_t = 0] + [\alpha_0 + \alpha_{1(s_t)} \varepsilon_{t-1}^2 + \beta_{1(s_t)} h_{t-1}] I[s_t = 1] \quad (15)$$

where $s = 0$ and $s = 1$ denote the low volatile and high volatile regimes, respectively.

3.5. Proposed hybrid model

The proposed hybrid model initially decomposes the time series into several components and then uses different techniques corresponding to the features of each component in order to provide a reliable prediction. Using CEEMD, the proposed hybrid model decomposes the time series into two categories: volatile and nonlinear series. Volatile series are forecasted by the MS-GARCH while the nonlinear series are forecasted by the SVM-PSO. Finally, by means of the linear addition of obtained results, the oil price is forecasted. The merit of this hybridization lies in the fact that it can cover various features of oil price time series. Fig. 1 illustrates the structure of the proposed hybrid model.

Following the proposed model is described step by step.

Step 1. Oil price time series is decomposed by the CEEMD. The oil price can be achieved as shown in Eq. (16).

$$X(t) = \sum_{i=1}^n x_i(t) \quad (16)$$

where x_i denotes oil price in series i .

Step 2. Time series decomposed by the CEEMD are subsumed under nonlinear and volatile components based on their characteristics. Therefore:

$$\sum_{i=1}^n x_i(t) = \sum_{i=1}^m N_i(t) + \sum_{i=m+1}^n V_i(t) \quad (17)$$

where N and V denote the nonlinear and volatile components, respectively. Generally, volatility can be described as an unpredictable and quick movement in the oil prices. Volatile components indicate that the price considerably goes up or down but makes retracement that could be seen as a reversal. In fact, in a volatile component, prices move very rapidly and tend to return to the area of previous support or resistance, while a non-volatile component does not have these features and steadily moves up or down with few stop points. Technically, volatility is used to refer to the extent of variability in the components with respect to the mean and standard deviation. Therefore, a volatile component is a series accompanied by much variability around a moving average, and a non-volatile component is a series accompanied by relatively low variability of the price. Basically, fluctuations in the oil price could relate to the inelasticity of supply and demand. In other words, crude oil prices are influenced by incidents which have a capacity to disrupt the flow of oil to the energy market like political conflicts, natural disasters, etc. Such incidents have the potential to cause disruptions or uncertainty about future supply or demand and, subsequently, severe volatilities in oil prices. Therefore, decomposing the time series into volatile and non-volatile portions is conducive to select a more reliable technique for the prediction.

Step 3. Components showing the volatile behavior of oil prices are forecasted by the MS-GARCH. Forecasted value is denoted by $\hat{V}_i(t)$. Volatility is considered as a conundrum which has a significant impact on the forecasting precision. Research shows that GARCH models can

effectively capture the volatility in time series [62]. However, recent studies report that oil price time series is accompanied by structural breaks or trend switching [63], and neglecting these characteristics may negatively affect the prediction accuracy [64]. An effective way to overcome this drawback in the oil price forecasting is to employ the MS-GARCH model, whose parameters can vary over time. This model can efficiently capture the variations in prices that boosts the forecasting quality.

Step 4. Nonlinear components are forecasted by the SVM-PSO. The author employs the PSO to set the parameters of SVM such as the penalty parameter and variance value in the kernel function. Forecasted value by the SVM-PSO is denoted by $\hat{N}_i(t)$. Research shows that using traditional models in forecasting nonlinear series generates large prediction errors [26] and using AI techniques is highly recommended. SVM-PSO is of the capacity to effectively solve the problem of small samples with high degree of nonlinearity, preclude local optimum, and overcome the shortcomings of traditional models [65].

Step 5. Forecasted values by both MS-GARCH and SVM-PSO are added linearly indicating the forecasted oil price.

$$\sum_{i=1}^n \hat{x}_i(t) = \sum_{i=1}^m \hat{N}_i(t) + \sum_{i=m+1}^n \hat{V}_i(t) \quad (18)$$

To assess the superiority of the proposed hybrid model, the author compares results with constituent algorithms, namely CEEMD-MS-GARCH and CEEMD-SVM-PSO. Also, I use the hybrid model proposed by Wu, Chen, Zhou and Li [18] as the benchmark and compare the result with that model as well. The benchmark is a hybrid model for oil price forecasting where, similar to my model, the authors initially use the CEEMD to decompose data. Next, ARIMA and sparse Bayesian learning with combined kernels are employed to forecast target values for each component. Then, the best forecast for each component is selected on the basis of the training accuracy. Finally, the selected values are used to forecast the oil price using a sparse Bayesian learning algorithm without kernel-tricks.

3.6. Performance evaluation

Model outcomes must be assessed in order to verify the validity of a forecasting [66]. In this way, various tests are executed to assess the forecasting performance of the proposed hybrid model and benchmark models.

- I. **MAE:** The MAE is a test applied to compare the forecasted values with realized data. This measure is the average of the absolute errors.

$$MAE = T^{-1} \sum_{t=1}^T |Y_{(t)} - \hat{Y}_{(t)}| \quad (19)$$

- II. **RMSE:** This test evaluates the variation between the forecasted values and realized data. That is to say, the RMSE indicates the

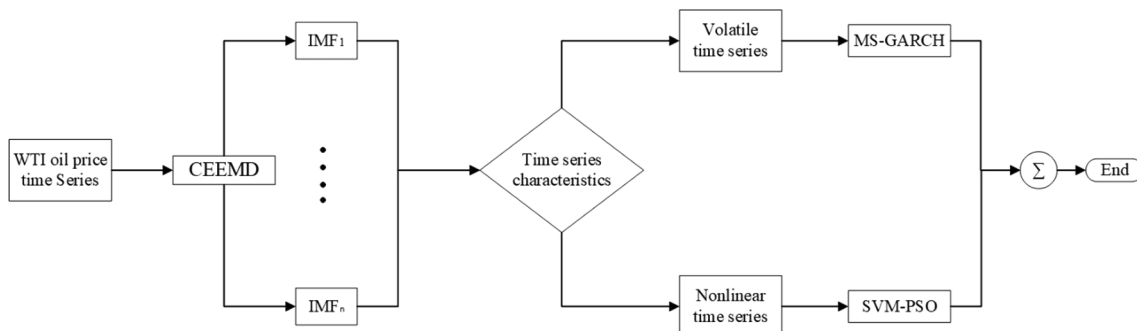


Fig. 1. Structure of the proposed hybrid model.

standard deviation of differences between forecasted and realized values. RMSE is known as an accuracy indicator used to compare forecasting errors of different models for a particular variable.

$$RMSE = \left(\frac{1}{T} \sum_{t=1}^T (Y_{(t)} - \hat{Y}_{(t)})^2 \right)^{1/2} \quad (20)$$

III. **MAPE:** MAPE is employed to measure the forecasting accuracy as a percentage.

$$MAPE = \frac{1}{T} \sum_{t=1}^T |Y_{(t)} - \hat{Y}_{(t)}|/Y_{(t)} \quad (21)$$

In above formulas T is the size of sample, $Y_{(t)}$ and $\hat{Y}_{(t)}$ are the actual and forecasted values at time t , respectively. Overall, when the values of these measures are smaller, the model prediction is more accurate.

Also, the DM test is a medium for making a comparison between the forecasting qualities in an attempt to ascertain if differences between competing forecasts are statistically significant. The DM test statistic is as follows:

$$DM = \frac{\bar{d}}{\sqrt{\frac{2\pi f_d}{T}}} \text{Normal}(0, 1) \quad (22)$$

where $\bar{d}$ is the mean of the loss differential between the two forecasts, T is the sample size, and $\hat{f}_d$ is a consistent estimate of spectral density of the loss differential.

4. Empirical results

4.1. Data

Fig. 2 illustrates the daily closing prices of WTI crude oil for the period from 6 June 2015 to 4 April 2016, giving a total of 252. The data were collected from Macrotrends website. The average WTI oil price in this period is \$45.13, the median is \$44.77 and the standard deviation is \$9.51. The vacillatory behavior of WTI oil prices underscores its forecasting importance in the market. It can be observed that the 47th observation indicates the highest price (\$61.43) occurred on 10 June 2015. Also, the 217th observation shows the least price (\$26.21) occurred on 11 February 2016. As can be seen in Fig. 2, the WTI oil price follows a decreasing trend overall. This trend could relate to the decrement in demands by large economies like China, India, Russia, and Brazil during the time period in question. Furthermore, another reason for the decline in prices was reportedly due to the oil supply, i.e. the increase in U.S. oil production resulted from advancements in efficient technologies which significantly reduced break-even costs for U.S. oil production. Also, geopolitical de-escalations, such as Iran nuclear deal or changes in OPEC policies, gave rise to an amicable situation which contributed to the price plunge. However, oil price sentiment caused many volatilities in the prices as it started rising at several points. As a

matter of fact, low oil prices gave rise to responsive policies designed by oil-exporter countries in order to increase oil prices. These countries have low levels of export diversification and their economies are quite dependent on petrodollars. The interplay between these factors caused a volatile behavior in the oil price time series. Generally, these interactions are a typical scenario throughout the last decades when it comes to oil prices and underscore the necessity of studying the time series in question. Also, WTI is one of the benchmarks in the global crude oil markets [67] and plays an important part in the global oil pricing mechanism. Since this time period shows diverse surges and behaviors, therefore, proposing a hybrid model to investigate this period may lead to a more reliable forecasting.

Literature reports various ways for dividing data into training and testing samples as it can have a prodigious impact on the performance of prediction. Forecasting validity is verified by measuring how well a model functions on a part of data that were not used for model fitting. That is to say, separation of the data into two parts, namely training and test data, is a standard practice upon which we use the training data to estimate forecasting parameters while the test data is utilized to examine the forecasting accuracy. Needless to say, a model with a good fitness for the training data will not necessarily generate a reliable prediction. Therefore, the forecasting performance must be examined using a part of the data that has not been used in the training process. In this study, 221 observations are used for the training set and the remaining 31 observations are used as testing data to forecast oil prices.

4.2. Forecasting results

To conduct the forecasting procedure, MATLAB and Eviews software are employed. The author initially decomposes the time series into nonlinear and volatile components using the CEEMD. Fig. 3 shows the IMFs derived from the decomposition. It can be observed that components $\{IMF_1, IMF_2, IMF_3\}$ contain volatile features of WTI oil prices while components $\{IMF_4, IMF_5, IMF_6, IMF_7, IMF_8\}$ encompass nonlinear features of prices.

The author employs the MS-GARCH to model and forecast the volatile series extracted by the CEEMD. First, the stationarity of IMFs should be verified because the effect of shocks on a stationary series will be disappeared gradually. In so doing, the author carries out the Augmented Dicky-Fuller (ADF) test. If the statistics of this test are significant, the author can conclude that the series are stationary. Table 2 presents the results of the ADF test and the stationarity of the IMFs is supported.

Autocorrelation coefficient is one of the criteria in modeling time series as it indicates the degree to which a time series is correlated with its past values. Hence, the Ljung-Box test is executed to ascertain the autocorrelation in the series. Table 3 shows the results for autocorrelation test.

In Table 3, the number of lags is calculated based on the ADF test. Also, take IMF_1 for instance, the Ljung-Box value for this component is

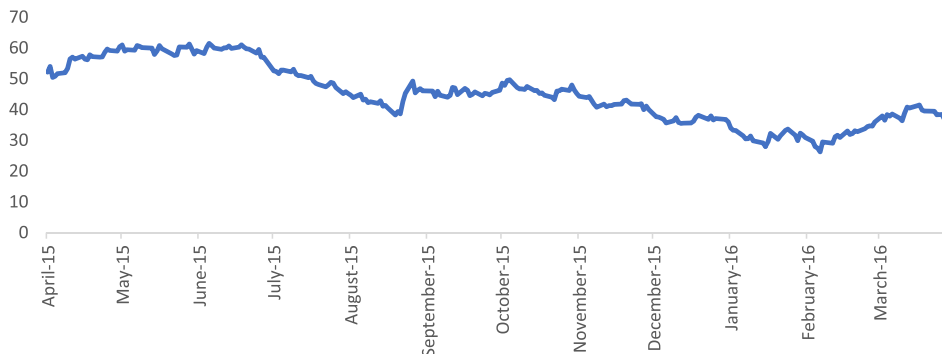


Fig. 2. Daily WTI oil prices (from June 2015 to April 2016).

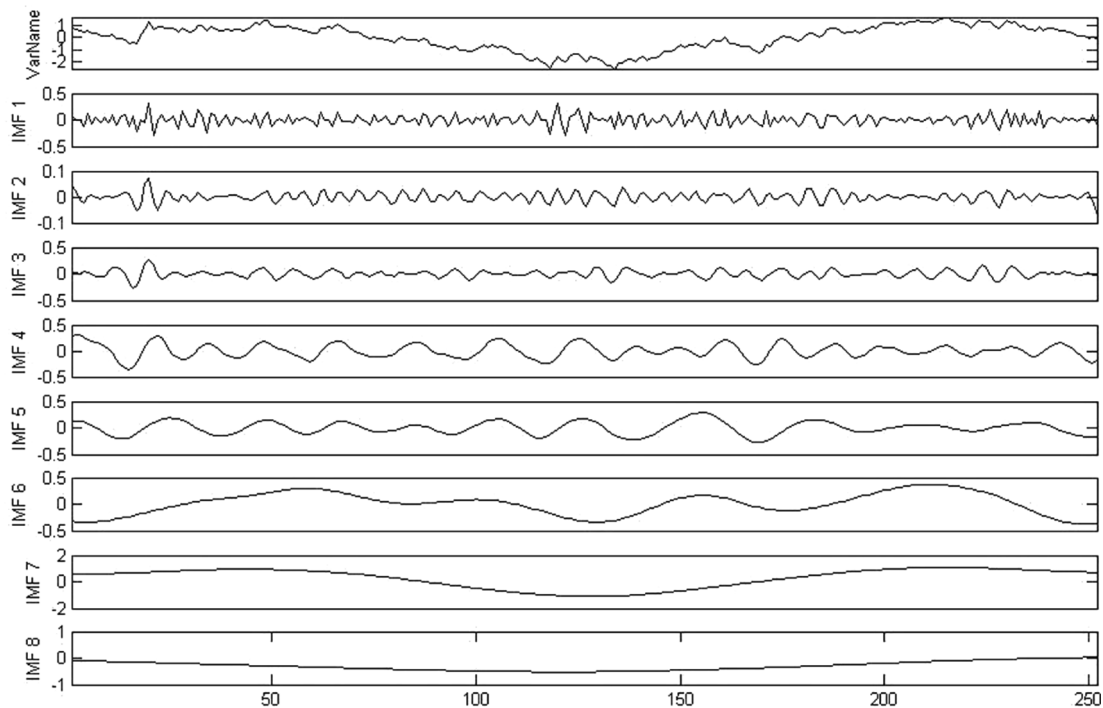


Fig. 3. Extracted IMFs of WTI daily prices by CEEMD.

Table 2
Results of the ADF unit root test.

Series	IMF ₁	IMF ₂	IMF ₃	IMF ₄	IMF ₅	IMF ₆	IMF ₇	IMF ₈
t statistic	-15.381	-12.7475	-10.1373	-8.0186	-4.7541	-2.9406	-4.8166	-37.8789

Table 3
Results of Ljung-Box test.

Series	IMF ₁	IMF ₂	IMF ₃	IMF ₄	IMF ₅	IMF ₆	IMF ₇	IMF ₈
Test statistic	62.977	214.15	521.31	384.86	6319.09	1113.01	761.29	249.56
Lag(s)	3	4	9	5	4	5	3	1

Table 4
Persistence of each components in regimes.

Series	IMF ₁	IMF ₂	IMF ₃	IMF ₄	IMF ₅	IMF ₆	IMF ₇	IMF ₈
Regime i	1.0675	1.058376	5.229876	1.019715	1.085367	1.164623	1.752868	6.477535
Regime ii	15.644	12.13942	1.328911	61.23123	12.77732	7.141621	2.327798	1.262930

Table 5
Values of regressions' variables.

Series	Regime	C	SIGMA	Series	Regime	C	SIGMA
IMF ₁	1	0.001634	-9.21965	IMF ₅	1	0.058582	-2.389186
	2	-0.000191	-2.314578		2	-0.121764	-2.628729
IMF ₂	1	-0.00255	-3.352773	IMF ₆	1	-0.3322947	-3.496512
	2	0.000156	-4.13694		2	0.056103	-1.783381
IMF ₃	1	0.002243	-2.896538	IMF ₇	1	-0.427193	-0.645525
	2	-0.004451	-2.23639		2	0.823697	-1.769126
IMF ₄	1	0.007498	-2.011181	IMF ₈	1	-0.489996	-3.718484
	2	-0.117255	-4.937634		2	-0.233757	-1.933289

76.471 which should be compared with the critical value ($\chi^2_{3,0.05} = 7.815$). As can be observed, the Ljung-Box value is greater than the critical value therefore, the null hypothesis implying the absence of

autocorrelation is rejected and the number of lags for this IMF is determined to be equal to 3. The next step is to forecast the volatile series using the MS-GARCH. For this purpose, I initially determine the type of

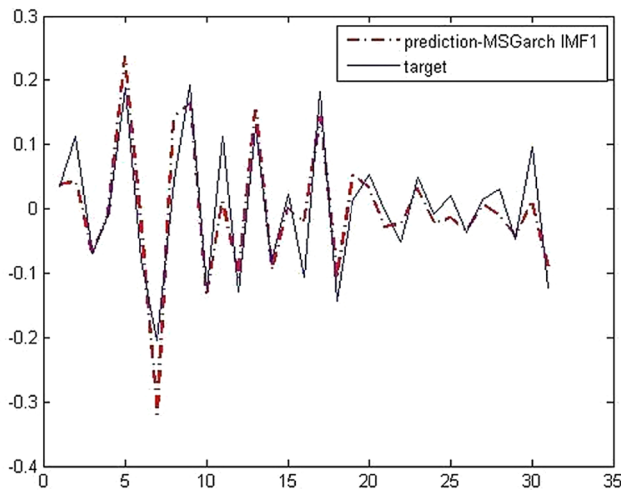


Fig. 4. Forecasted results for IMF₁ by MS-GARCH versus actual data.

regimes existing in each component to adopt the proper forecasting policy. Each IMF extracted by the CEEMD may be of various regimes; however, the author defines two classes as follows.

- (i). Low-volatile regime: IMFs which last in low-volatile regime for a longer period of time and mostly are plotted in the low-volatile regime during the time period of the study (The fraction of their low-volatile regime to high-volatile regime is greater than 1.5).
- (ii). High-volatile regime: IMFs which last in high-volatile regime for a longer period of time and mostly are plotted in the high-volatile regime during the time period of the study (The fraction of their low-volatile regime to high-volatile regime is less than 0.5).

Assuming each of the regimes as a state, the transition probabilities can be calculated. Accordingly, the time length of persistence in each state can be found by using the matrix of transition probabilities. Taking into consideration the duration that a series remains in each state, we can determine the regime which is more suitable for forecasting. To do this, the author chooses the regime which lasts longer. The persistence time in each regime is presented in Table 4. Before going any further, the significance of the selected regime for forecasting should be tested. Table 5 presents the results for regressions' variables.

The proper regime for each IMF is selected according to the results

Table 6

Optimal parameters of SVM using PSO.

Series	C	ϵ	γ	Series	C	ϵ	γ
IMF ₁	1.52	0.381	0.59	IMF ₁	1.67	0.782	0.82
IMF ₂	1.39	0.723	0.51	IMF ₂	1.52	0.0453	0.73
IMF ₃	1.43	0.716	0.53	IMF ₃	1.48	0.0368	0.76
IMF ₄	1.59	0.141	0.68	IMF ₄	1.46	0.0353	0.71

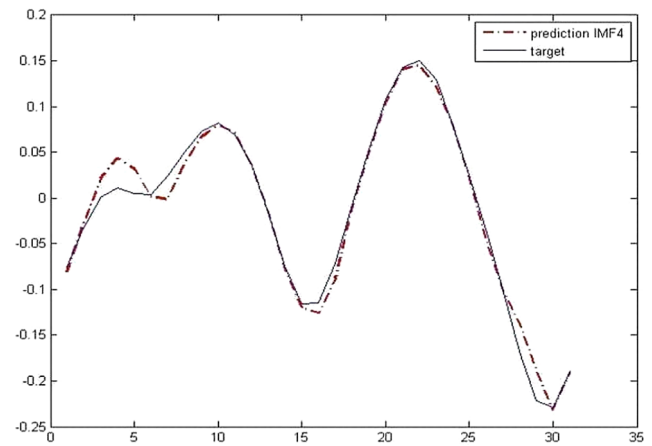


Fig. 6. Forecasted results for IMF₄ by SVM-PSO versus actual data.

presented in Table 4. For instance, the author selects regime *ii* for IMF₁ where the time series lasts in regime *ii* for a longer time compared to the time lasting in regime *i*. Fig. 4. Shows the forecasting result for IMF₁. Also, the forecasting results for all IMFs are shown in Fig. 5. As can be observed, the volatile behavior of IMF₃ has been forecasted with a proper accuracy but the forecasting results for series IMF₁ and IMF₂ have lower accuracy. One reason lies in the fact that IMF₃ shows less volatile behavior and MS-GARCH is able to suitably capture these volatilities, while IMF₁ and IMF₂ show a higher volatility which leads to a decline in the accuracy of the MS-GARCH. Thus, the difference between actual values and forecasting values would be higher.

The next step is to forecast the extracted components using the SVM-PSO. As mentioned in Section 3.5, the author utilizes PSO to estimate the parameters of SVM. Table 6 presents the optimal set of the SVM parameters for all IMFs. This algorithm indicates high accuracy in forecasting nonlinear features existing in the time series due to its

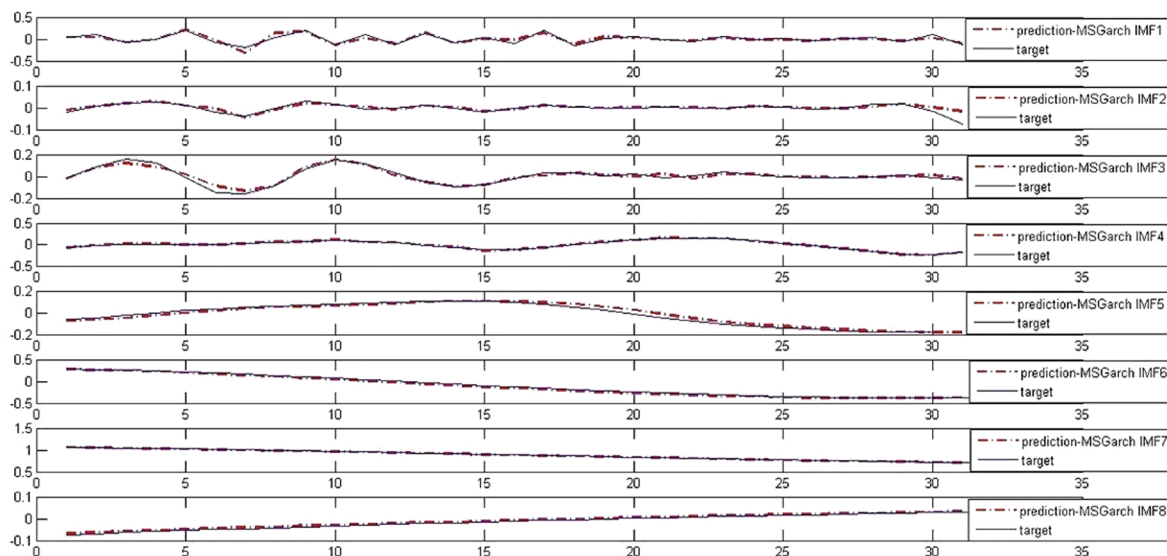


Fig. 5. Forecasted results for IMFs by MS-GARCH versus actual data.

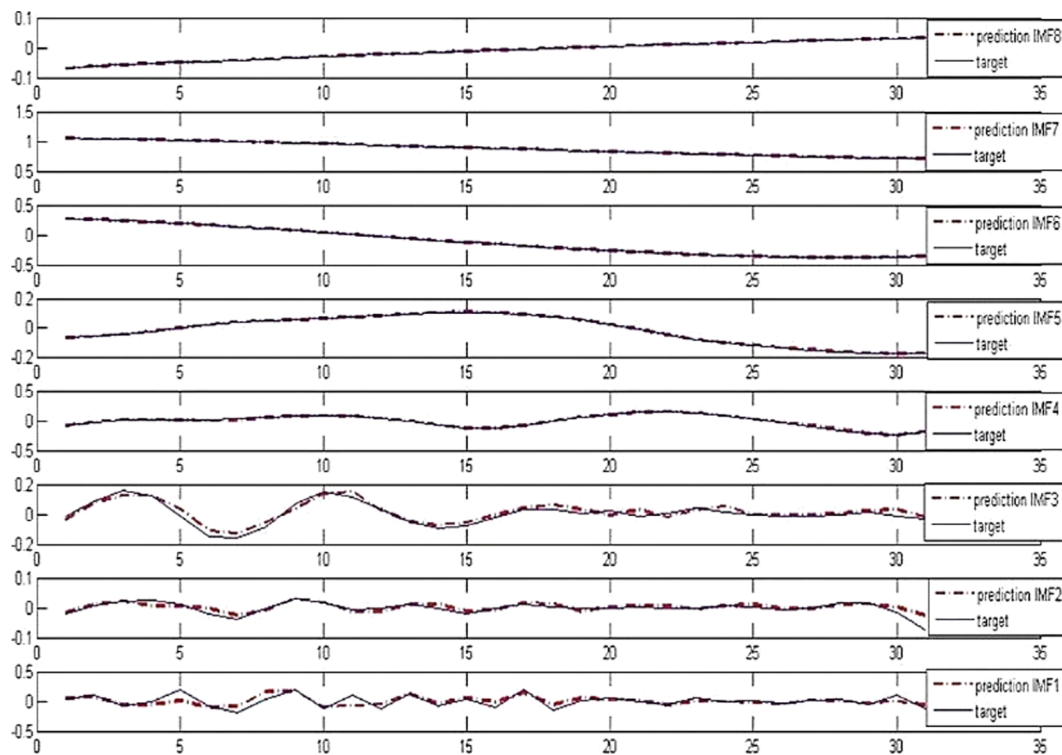


Fig. 7. Forecasted results for IMFs by SVM-PSO versus actual data.

Table 7
Errors for forecasting volatile and nonlinear IMFs.

Model	MAE	RMSE	MAPE
Volatile components (IMF ₁ , IMF ₂ , IMF ₃)			
SVM-PSO	0.049	0.066	0.087
MS-GARCH	0.042	0.055	0.076
Nonlinear components (IMF ₄ , IMF ₅ , IMF ₆ , IMF ₇ , IMF ₈)			
SVM-PSO	0.043	0.059	0.081
MS-GARCH	0.051	0.065	0.090

strong fundamental theory. Using quadratic programming enables the SVM to achieve a global solution, be less likely to overestimate, and reach a geometric interpretation. Fig. 6 shows the forecasting result for one of the IMFs. Also, forecasting results for all IMFs are presented in Fig. 7.

To forecast the oil price by the proposed hybrid model, the extracted characteristics decomposed by the CEEMD are classified into two groups. First group indicates the volatile feature of price ({IMF₁, IMF₂, IMF₃}) and the second group shows nonlinear feature ({IMF₄, IMF₅, IMF₆, IMF₇, IMF₈}). According to the results presented in Table 7, the MS-GARCH model forecasts volatile components with lower errors while the SVM-PSO produces smaller errors when it comes to forecasting nonlinear feature. Thus, this hybridization results in a more accurate prediction compared to those of the constituent models. Fig. 8 illustrates the forecasted results for all IMFs using the proposed hybrid model.

Table 8 represents comparisons between models for forecasting the 31 final observations. In this case, the errors for the benchmark model is also presented. Considering the obtained values for the RMSE, MAE, and MAPE, it can be concluded that the proposed hybrid model outperforms the counterparts. This could relate to the fact that volatility and structural breaks are two important factors for the accuracy of time series prediction and the proposed hybrid model is able to capture them efficiently. The author also performs the DM test to evaluate forecasting quality of the hybrid model in a pairwise comparison with other

counterparts. Null hypothesis of the DM test implies that there is no significant difference between errors if the DM statistics are less than 1.96. Greater values for the DM reject the null hypothesis. According to Table 9, $|DM| \geq 1.96$ is confirmed only for the proposed hybrid model. That is to say, there is a significant difference between errors when we use the proposed hybrid model compared with other models. Thus, using the proposed hybrid model eventuates in the paramount forecast.

A significant challenge about financial forecasting is the very disclaimer that “past performance is no guarantee for the future”. Although this is the case in portfolio formulation in financial markets, the crude oil can be subsumed under the category of energy commodities offered in the energy market. Owing to the fact that changes in the oil price have always been due to some geopolitical factors, natural disasters, and interactions between supply and demand; therefore, studying previous behavior allows us to recognize the patterns of oil prices behavior. It is axiomatic that what has happened erstwhile is of the potential to reoccur with a modified structure. Accordingly, we already know that there must be shifts in oil prices whenever we confront with such incidents. Hence, forecasting provides us with a potential tool beneficial to direct us towards a more reasonable, but not necessarily true, decision making in the future. Moreover, we need some facts to conduct a robust analysis and oil price is no exception. By and large, this study is tantamount to an attempt to improve the prediction accuracy in the literature.

5. Conclusion

Oil price complexity could relate to the fact that the corresponding time series includes nonlinearity, structural breaks, and lag. In this paper, the author proposes a hybrid model for oil price forecasting whose focus is on capturing these features in order to enhance the prediction accuracy. For this purpose, the author initially decomposes the time series into several components using the complete ensemble empirical mode decomposition and then employs different forecasting techniques corresponding to the characteristics of each component in order to achieve a reliable prediction. Overall, I divided the extracted components into

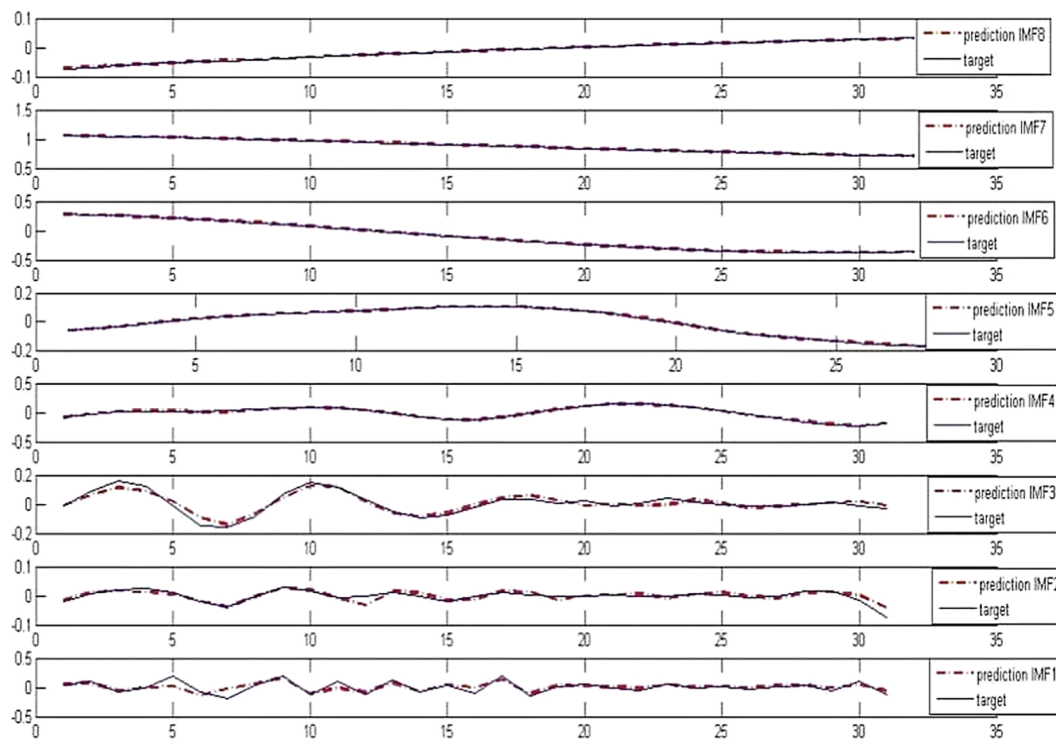


Fig. 8. Forecasted results by proposed hybrid model and actual values.

Table 8
The errors of different forecasting models.

Model	MAE	RSME	MAPE
SVM-PSO	0.047	0.061	0.061
MS-GARCH	0.048	0.06	0.065
Proposed hybrid model	0.042	0.053	0.057
Benchmark model	0.044	0.057	0.060

Table 9
Results of the DM test.

Pair comparison	RMSE	MAE	MAPE
Hybrid & SVM-PSO	−2.8264	−3.4016	−2.5602
Hybrid & MS-GARCH	−5.4001	−3.2937	−4.1216
Hybrid & Benchmark	−1.9802	−2.0081	−1.9737

volatile and nonlinear classes. This is conducive to employ the appropriate forecasting techniques which eventually leads to a more promising prediction. To forecast the volatile components, I applied the Markov-switching-generalized autoregressive conditional heteroskedasticity because of its capacity to capture persistence and regime switch. According to the results, this model predicted the volatile components with a proper accuracy and had the best performance in forecasting the component with low volatility. However, nonlinear components were forecasted by the support vector machine trained by particle swarm optimization where I utilized this optimization algorithm in order to find the optimal parameters for the support vector machine. According to the results, this algorithm turned out to be an effective tool for forecasting nonlinear components due to its capacity to achieve a global solution. Finally, the linear addition of both forecasted nonlinear components and volatile components gave us a reliable forecast of the oil price. The author used daily closing price of West Texas Intermediate crude oil to verify the proposed model. Root mean square error, mean absolute error, and mean absolute percentage error were used in computing forecasting errors where the results demonstrated the superiority of the proposed hybrid model over

the counterparts. Also, the results of pairwise comparisons by the Diebold-Mariano test documented that only the proposed hybrid model leads to quality forecasting values.

The core objective of the paper was to offer some improvements on the forecasting accuracy with respect to the pertinent literature. According to the empirical analysis, the present study was successful in achieving this goal as the proposed hybrid model outperforms its counterparts. Undoubtedly, proposing a specific model which can lead to a precise forecasting is almost impossible. However, such forecasting is tantamount to an effort in order to depict a frame of the future by which we can be prepared to adopt an effective policy, depending on the exigency of the circumstances. The proposed hybrid model may be further developed from the following perspectives. Combination of the support vector machine with other methods such as ant colony optimization algorithm or genetic algorithm can be investigated. Also, the proposed model needs to be applied to forecast other energy commodities in order to find its robustness and generalizability. The author addresses these points in not too distant future.

CRediT authorship contribution statement

Hooman Abdollahi: Conceptualization, Methodology, Data curation, Software, Formal analysis, Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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